Examiner only

4. In a class experiment, Dylan and Joel were given a solution of sodium carbonate containing 5.23 g of sodium carbonate powder dissolved in 500 cm^3 of water.

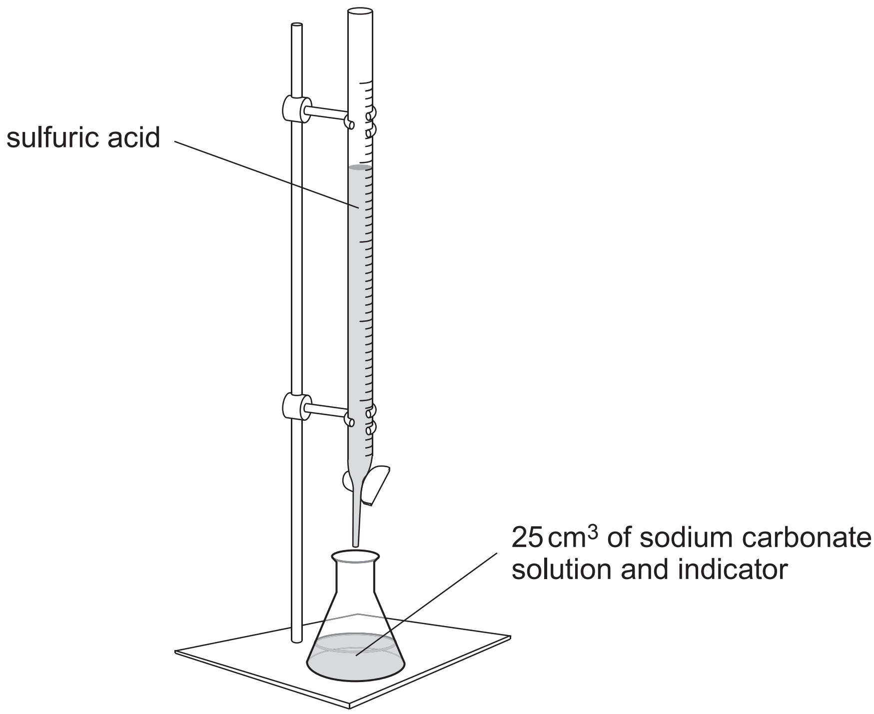
The relative formula mass (M_r) of sodium carbonate is 106.

(a) Use this information to calculate the number of moles of sodium carbonate in the 5.23 g of the powder. Give your answer correct to **two** decimal places. [2]

Number of moles = ... mol

(b) Dylan and Joel were asked to use the sodium carbonate solution to prepare a sample of sodium sulfate crystals in a three-stage process.

In the first stage of their preparation, they used the following apparatus to carry out a titration.



The equation for the reaction taking place is as follows.

$$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2$$

 (3430UE0-1)

Examiner only

(i) A trial run was carried out and the titration repeated three times. The volume of acid added each time was recorded.

	Trial	1	2	3
Volume of sulfuric acid added (cm^3)	30.20	27.55	27.75	27.65

I. State the purpose of carrying out a trial run. [1]

....

....

II. State whether the sulfuric acid or the sodium carbonate solution is the more concentrated. Give the reason for your answer. [1]

....

....

(ii) Use **all** of the information provided to describe **in detail** the other two stages Dylan and Joel carried out to obtain pure sodium sulfate crystals. [4]

....

....

....

....

....

....

....

....

Examiner only

(c) Sodium sulfate solution is also formed when sodium hydroxide solution reacts with copper(II) sulfate solution.

(i) Give the balanced symbol equation for this reaction. [3]

....

(ii) Describe tests that can be carried out to identify both of the ions in sodium sulfate solution. Give the expected observation for both tests. [2]

....

....

....

....

(d) The reaction occurring between solutions of sodium carbonate and magnesium sulfate forms a precipitate of magnesium carbonate.

Write the **ionic** equation for the formation of magnesium carbonate. [2]

.... + ... $\longrightarrow$ ....

15